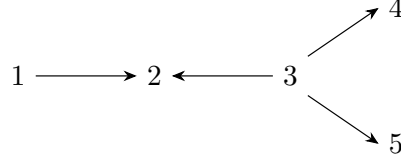


Final Exam - Applied Abstract Algebra

June 9th, 2026

First Problem Variants

Variant 1. (40 points) Let Q be a quiver:



- Construct $P(3)$ and $I(2)$ and using projective and injective resolution calculate $\tau(P(3))$ and $\tau^{-1}(I(2))$, where τ denotes the Auslander-Reiten translation.
- Construct a triangulation of punctured polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(3)$, $I(2)$, $\tau(P(3))$, $\tau^{-1}(I(2))$ and to indecomposable representation of dimension vector $(1, 2, 2, 1, 1)$.

Proof. The projective $P(i)$ is spanned by paths starting at i , and the injective $I(i)$ is spanned by paths ending at i . Hence

$$\dim P(3) = (0, 1, 1, 1, 1), \quad \dim I(2) = (1, 1, 1, 0, 0).$$

The nonzero maps in $P(3)$ are the identity maps on the arrows $3 \rightarrow 2$, $3 \rightarrow 4$, and $3 \rightarrow 5$; the nonzero maps in $I(2)$ are the identity maps on $1 \rightarrow 2$ and $3 \rightarrow 2$.

Since $P(3)$ is projective, its minimal projective resolution is $0 \rightarrow 0 \rightarrow P(3) \rightarrow P(3) \rightarrow 0$, so applying the Nakayama definition of Auslander-Reiten translation gives $\tau P(3) = 0$. Similarly, $I(2)$ is injective, so $\tau^{-1} I(2) = 0$.

Let $T_Q = \{\gamma_1, \dots, \gamma_5\}$ be the triangulation of the punctured pentagon obtained from Q by Schiffler's construction, with γ_i corresponding to the vertex i . Let R denote elementary clockwise rotation, with tag change at the puncture. Then

$$P(3) \leftrightarrow R^{-1}\gamma_3, \quad I(2) \leftrightarrow R\gamma_2.$$

The actual objects $\tau P(3)$ and $\tau^{-1} I(2)$ are zero in $\text{rep } Q$; formally rotating the arcs gives the boundary triangulation arcs γ_3 and γ_2 , respectively.

The indecomposable representation of dimension vector $(1, 2, 2, 1, 1)$ may be written as

$$X_1 = X_4 = X_5 = k, \quad X_2 = X_3 = k^2,$$

with arrow maps

$$X_{1 \rightarrow 2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad X_{3 \rightarrow 2} = I_2, \quad X_{3 \rightarrow 4} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad X_{3 \rightarrow 5} = \begin{pmatrix} 0 & 1 \end{pmatrix}.$$

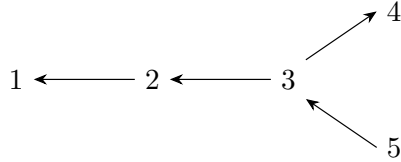
If f is an endomorphism of X , then the two maps out of X_3 force f_3 to be diagonal, the identity map $X_3 \rightarrow X_2$ gives $f_2 = f_3$, and the map $X_1 \rightarrow X_2$ forces the two diagonal

entries to be equal. Thus $\text{End}(X) = k$, so X is indecomposable. Geometrically, X corresponds to the unique arc ρ not in T_Q such that

$$(e(\rho, \gamma_1), \dots, e(\rho, \gamma_5)) = (1, 2, 2, 1, 1).$$

□

Variant 2. (40 points) Let Q be a quiver:



- Construct $P(3)$ and $I(2)$ and using projective and injective resolution calculate $\tau(P(3))$ and $\tau^{-1}(I(2))$, where τ denotes the Auslander-Reiten translation.
- Construct a triangulation of punctured polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(3)$, $I(2)$, $\tau(P(3))$, $\tau^{-1}(I(2))$ and to indecomposable representation of dimension vector $(1, 2, 2, 1, 1)$.

Proof. By the path description of projectives and injectives,

$$\dim P(3) = (1, 1, 1, 1, 0), \quad \dim I(2) = (0, 1, 1, 0, 1).$$

The nonzero maps in $P(3)$ are identities along $3 \rightarrow 2$, $2 \rightarrow 1$, and $3 \rightarrow 4$; the nonzero maps in $I(2)$ are identities along $5 \rightarrow 3$ and $3 \rightarrow 2$. Since $P(3)$ is projective and $I(2)$ is injective,

$$\tau P(3) = 0, \quad \tau^{-1} I(2) = 0.$$

Construct the punctured-pentagon triangulation $T_Q = \{\gamma_1, \dots, \gamma_5\}$ by Schiffler's rule, labeling γ_i by the vertex i . With R the clockwise tagged rotation,

$$P(3) \leftrightarrow R^{-1}\gamma_3, \quad I(2) \leftrightarrow R\gamma_2.$$

Thus the formal rotated arcs for $\tau P(3)$ and $\tau^{-1} I(2)$ are γ_3 and γ_2 , while the actual representations are zero.

For the dimension vector $(1, 2, 2, 1, 1)$ take $X_1 = X_4 = X_5 = k$ and $X_2 = X_3 = k^2$, with maps

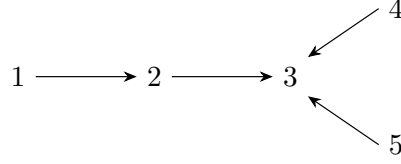
$$X_{2 \rightarrow 1} = \begin{pmatrix} 1 & -1 \end{pmatrix}, \quad X_{3 \rightarrow 2} = I_2, \quad X_{3 \rightarrow 4} = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad X_{5 \rightarrow 3} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The endomorphism relations force the common matrix on X_2 and X_3 to be diagonal, and then the maps to X_1 and X_4 force the two diagonal entries to agree. Hence $\text{End}(X) = k$, so X is indecomposable. It is the arc ρ with crossing vector

$$(e(\rho, \gamma_1), \dots, e(\rho, \gamma_5)) = (1, 2, 2, 1, 1).$$

□

Variant 3. (40 points) Let Q be a quiver:



- Construct $P(2)$ and $I(3)$ and using projective and injective resolution calculate $\tau(P(2))$ and $\tau^{-1}(I(3))$, where τ denotes the Auslander-Reiten translation.
- Construct a triangulation of punctured polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(2)$, $I(3)$, $\tau(P(2))$, $\tau^{-1}(I(3))$ and to indecomposable representation of dimension vector $(1, 2, 2, 1, 1)$.

Proof. The projective and injective representations are

$$\dim P(2) = (0, 1, 1, 0, 0), \quad \dim I(3) = (1, 1, 1, 1, 1).$$

In $P(2)$ the only nonzero arrow map is the identity on $2 \rightarrow 3$; in $I(3)$ all four arrow maps are identities. Since $P(2)$ is projective and $I(3)$ is injective,

$$\tau P(2) = 0, \quad \tau^{-1} I(3) = 0.$$

Let $T_Q = \{\gamma_1, \dots, \gamma_5\}$ be the punctured-pentagon triangulation associated to this orientation. Then

$$P(2) \leftrightarrow R^{-1}\gamma_2, \quad I(3) \leftrightarrow R\gamma_3.$$

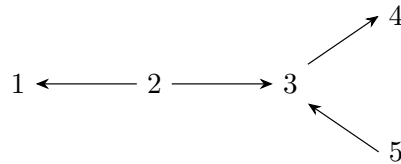
The formal arcs obtained by rotating once are γ_2 and γ_3 ; these are triangulation arcs, while the corresponding translated representations in $\text{rep } Q$ are zero.

The required indecomposable of dimension vector $(1, 2, 2, 1, 1)$ is obtained by taking $X_1 = X_4 = X_5 = k$, $X_2 = X_3 = k^2$, and

$$X_{1 \rightarrow 2} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad X_{2 \rightarrow 3} = I_2, \quad X_{4 \rightarrow 3} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad X_{5 \rightarrow 3} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The two maps into X_3 make the common endomorphism of X_2 and X_3 diagonal in the standard basis, and the map from X_1 forces the two diagonal entries to be equal. Therefore $\text{End}(X) = k$, so X is indecomposable. It corresponds to the unique arc ρ with crossing vector $(1, 2, 2, 1, 1)$ against the arcs γ_i . $\square$

Variant 4. (40 points) Let Q be a quiver:



- Construct $P(2)$ and $I(3)$ and using projective and injective resolution calculate $\tau(P(2))$ and $\tau^{-1}(I(3))$, where τ denotes the Auslander-Reiten translation.

- b). Construct a triangulation of punctured polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(2)$, $I(3)$, $\tau(P(2))$, $\tau^{-1}(I(3))$ and to indecomposable representation of dimension vector $(1, 2, 2, 1, 1)$.

Proof. The path descriptions give

$$\dim P(2) = (1, 1, 1, 1, 0), \quad \dim I(3) = (0, 1, 1, 0, 1).$$

The nonzero maps in $P(2)$ are identities along $2 \rightarrow 1$, $2 \rightarrow 3$, and $3 \rightarrow 4$; the nonzero maps in $I(3)$ are identities along $2 \rightarrow 3$ and $5 \rightarrow 3$. Hence

$$\tau P(2) = 0, \quad \tau^{-1} I(3) = 0.$$

If $T_Q = \{\gamma_1, \dots, \gamma_5\}$ is the corresponding triangulation of the punctured pentagon and R is clockwise tagged rotation, then

$$P(2) \leftrightarrow R^{-1}\gamma_2, \quad I(3) \leftrightarrow R\gamma_3.$$

Thus the formal arcs for the two translated terms are γ_2 and γ_3 , although the actual representations are zero.

For the vector $(1, 2, 2, 1, 1)$ take $X_1 = X_4 = X_5 = k$ and $X_2 = X_3 = k^2$ with

$$X_{2 \rightarrow 1} = (1 \ -1), \quad X_{2 \rightarrow 3} = I_2, \quad X_{3 \rightarrow 4} = (1 \ 0), \quad X_{5 \rightarrow 3} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The same endomorphism calculation as above gives $\text{End}(X) = k$, so X is indecomposable. In the punctured pentagon it is the unique arc ρ satisfying

$$(e(\rho, \gamma_1), \dots, e(\rho, \gamma_5)) = (1, 2, 2, 1, 1).$$

□

Variant 5. (40 points) Let Q be a quiver:

$$1 \longleftarrow 2 \longleftarrow 3 \longrightarrow 4 \longleftarrow 5$$

- a). Compute the Auslander-Reiten quiver for Q . Calculate $\text{Hom}(P(3), I(4))$.
- b). Construct a triangulation of polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(3)$, $I(4)$, $\tau(P(3))$, $\tau^{-1}(I(4))$ and to indecomposable representation of dimension vector $(1, 1, 1, 1, 1)$.

Proof. Write X_{ab} for the interval representation supported on $a, a+1, \dots, b$. The vertices of the Auslander-Reiten quiver are all X_{ab} with $1 \leq a \leq b \leq 5$, and its irreducible arrows are

$$\begin{aligned} X_{11} &\rightarrow X_{12}, & X_{12} &\rightarrow X_{14}, X_{22}, & X_{13} &\rightarrow X_{23}, \\ X_{14} &\rightarrow X_{15}, X_{24}, & X_{15} &\rightarrow X_{13}, X_{25}, & X_{22} &\rightarrow X_{24}, \\ X_{23} &\rightarrow X_{33}, & X_{24} &\rightarrow X_{25}, X_{34}, & X_{25} &\rightarrow X_{23}, X_{35}, \\ X_{34} &\rightarrow X_{35}, & X_{35} &\rightarrow X_{33}, X_{55}, & X_{44} &\rightarrow X_{14}, X_{45}, & X_{45} &\rightarrow X_{15}. \end{aligned}$$

Here $P(3) = X_{14}$ and $I(4) = X_{35}$. Since $\text{Hom}(P(i), M) \cong M_i$ naturally,

$$\text{Hom}(P(3), I(4)) \cong I(4)_3 \cong k,$$

so $\dim \text{Hom}(P(3), I(4)) = 1$.

Number the vertices of an 8-gon clockwise. A triangulation corresponding to Q is

$$T_1 = (1, 3), \quad T_2 = (3, 8), \quad T_3 = (3, 7), \quad T_4 = (4, 7), \quad T_5 = (4, 6).$$

A diagonal not in the triangulation represents the interval whose support is exactly the set of triangulation diagonals that it crosses. Thus

$$P(3) = X_{14} \leftrightarrow (2, 6), \quad I(4) = X_{35} \leftrightarrow (5, 8), \quad X_{15} \leftrightarrow (2, 5).$$

In $\text{rep } Q$, $\tau P(3) = 0$ and $\tau^{-1}I(4) = 0$. Under the formal polygon rotation, these boundary positions are the triangulation diagonals $T_3 = (3, 7)$ and $T_4 = (4, 7)$, respectively. $\square$

Variant 6. (40 points) Let Q be a quiver:

$$1 \longrightarrow 2 \longrightarrow 3 \longleftarrow 4 \longleftarrow 5$$

- Compute the Auslander-Reiten quiver for Q . Calculate $\text{Hom}(P(1), I(3))$.
- Construct a triangulation of polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(1)$, $I(3)$, $\tau(P(1))$, $\tau^{-1}(I(3))$ and to indecomposable representation of dimension vector $(1, 1, 1, 1, 1)$.

Proof. Again let X_{ab} denote the interval representation supported on $a, \dots, b$. The Auslander-Reiten quiver has vertices X_{ab} , $1 \leq a \leq b \leq 5$, and irreducible arrows

$$\begin{aligned} X_{12} &\rightarrow X_{11}, & X_{13} &\rightarrow X_{14}, & X_{14} &\rightarrow X_{15}, X_{44}, & X_{15} &\rightarrow X_{12}, X_{45}, \\ X_{22} &\rightarrow X_{12}, & X_{23} &\rightarrow X_{13}, X_{24}, & X_{24} &\rightarrow X_{14}, X_{25}, \\ X_{25} &\rightarrow X_{15}, X_{22}, & X_{33} &\rightarrow X_{23}, X_{34}, \\ X_{34} &\rightarrow X_{24}, X_{35}, & X_{35} &\rightarrow X_{25}, & X_{44} &\rightarrow X_{45}, & X_{45} &\rightarrow X_{55}. \end{aligned}$$

We have $P(1) = X_{13}$ and $I(3) = X_{15}$. Therefore

$$\text{Hom}(P(1), I(3)) \cong I(3)_1 \cong k,$$

and the Hom space has dimension 1.

A corresponding triangulation of the clockwise numbered 8-gon is

$$T_1 = (1, 3), \quad T_2 = (1, 4), \quad T_3 = (1, 5), \quad T_4 = (5, 8), \quad T_5 = (5, 7).$$

The relevant diagonals are

$$P(1) = X_{13} \leftrightarrow (2, 8), \quad I(3) = X_{15} \leftrightarrow (2, 6), \quad X_{15} \leftrightarrow (2, 6).$$

The actual translated representations are $\tau P(1) = 0$ and $\tau^{-1}I(3) = 0$; the formal rotated arcs are $T_1 = (1, 3)$ and $T_3 = (1, 5)$. $\square$

Variant 7. (40 points) Let Q be a quiver:

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow 4 \longleftarrow 5$$

- Compute the Auslander-Reiten quiver for Q . Calculate $\text{Hom}(P(1), I(4))$.
- Construct a triangulation of polygon corresponding to Q and find which arcs with respect to this triangulation correspond to $P(1)$, $I(4)$, $\tau(P(1))$, $\tau^{-1}(I(4))$ and to indecomposable representation of dimension vector $(1, 1, 1, 1, 1)$.

Proof. Let X_{ab} be the interval representation supported on $a, \dots, b$. The Auslander-Reiten quiver has irreducible arrows

$$\begin{aligned} X_{12} &\rightarrow X_{11}, & X_{13} &\rightarrow X_{12}, & X_{14} &\rightarrow X_{15}, & X_{15} &\rightarrow X_{13}, X_{55}, \\ X_{22} &\rightarrow X_{12}, & X_{23} &\rightarrow X_{13}, X_{22}, & X_{24} &\rightarrow X_{14}, X_{25}, \\ X_{25} &\rightarrow X_{15}, X_{23}, & X_{33} &\rightarrow X_{23}, & X_{34} &\rightarrow X_{24}, X_{35}, \\ X_{35} &\rightarrow X_{25}, X_{33}, & X_{44} &\rightarrow X_{34}, X_{45}, & X_{45} &\rightarrow X_{35}. \end{aligned}$$

Here $P(1) = X_{14}$ and $I(4) = X_{15}$, so

$$\text{Hom}(P(1), I(4)) \cong I(4)_1 \cong k.$$

Thus $\dim \text{Hom}(P(1), I(4)) = 1$.

One triangulation of the clockwise numbered 8-gon corresponding to this orientation is

$$T_1 = (1, 3), \quad T_2 = (1, 4), \quad T_3 = (1, 5), \quad T_4 = (1, 6), \quad T_5 = (6, 8).$$

The relevant diagonals are

$$P(1) = X_{14} \leftrightarrow (2, 8), \quad I(4) = X_{15} \leftrightarrow (2, 7), \quad X_{15} \leftrightarrow (2, 7).$$

The actual translated representations are zero: $\tau P(1) = 0$ and $\tau^{-1} I(4) = 0$. The formal polygon rotations land on the triangulation diagonals $T_1 = (1, 3)$ and $T_4 = (1, 6)$. $\square$

Common Problems

Choose up to three problems from this list to submit.

- (20 points) Prove that P is projective if and only if $\text{Ext}^1(P, N) = 0$ for all representations N .

Proof. If P is projective, every short exact sequence

$$0 \rightarrow N \rightarrow E \rightarrow P \rightarrow 0$$

splits, since the identity map of P lifts through $E \rightarrow P$. Hence every extension of P by N is trivial, so $\text{Ext}^1(P, N) = 0$.

Conversely, assume that $\text{Ext}^1(P, N) = 0$ for every N . Let $g : M \rightarrow X$ be an epimorphism and let $f : P \rightarrow X$ be a map. Form the pullback

$$E = M \times_X P = \{(m, p) \mid g(m) = f(p)\}.$$

Then

$$0 \rightarrow \ker g \rightarrow E \rightarrow P \rightarrow 0$$

is exact. Its class lies in $\text{Ext}^1(P, \ker g) = 0$, so it splits. A section $P \rightarrow E$, followed by the projection $E \rightarrow M$, gives a lift of f through g . Thus $\text{Hom}(P, -)$ sends epimorphisms to epimorphisms, and P is projective. $\square$

2. (20 points) Let $M \in \text{rep } Q$ and

$$0 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

be standard projective resolution. Show that

$$0 \rightarrow D(M) \rightarrow D(P_0) \rightarrow D(P_1) \rightarrow 0$$

is the injective resolution for $D(M) \in \text{rep } Q^{\text{op}}$. Show that duality functor D takes a minimal projective resolution for M to a injective envelope for $D(M)$.

Proof. Write $D = \text{Hom}_k(-, k)$. Since all vector spaces are finite-dimensional, D is exact and contravariant. Applying D to

$$0 \rightarrow P_1 \rightarrow P_0 \rightarrow M \rightarrow 0$$

gives an exact sequence

$$0 \rightarrow D(M) \rightarrow D(P_0) \rightarrow D(P_1) \rightarrow 0.$$

For every vertex i , the dual of the projective $P_Q(i)$ is the injective $I_{Q^{\text{op}}}(i)$: a path $i \rightarrow j$ in Q is the same as a path $j \rightarrow i$ in Q^{op} after reversing arrows and dualizing. Hence $D(P_0)$ and $D(P_1)$ are injective representations of Q^{op} . Therefore the displayed sequence is an injective resolution of $D(M)$.

If the projective resolution is minimal, then the map $P_0 \rightarrow M$ is a projective cover. Its dual $D(M) \hookrightarrow D(P_0)$ is an essential monomorphism into an injective object: equivalently, if $a : D(P_0) \rightarrow D(P_0)$ satisfies $a|_{D(M)} = \text{id}_{D(M)}$, then dualizing gives an endomorphism of P_0 over M , hence an automorphism by minimality of the projective cover. Thus $D(M) \hookrightarrow D(P_0)$ is an injective envelope, and the dual complex is the minimal injective resolution. $\square$

3. (20 points) Let $g : M \rightarrow N$ be a surjective morphism between representations of Q , and let $P(i)$ be the projective representation at vertex i . Show that the map

$$g_* : \text{Hom}(P(i), M) \rightarrow \text{Hom}(P(i), N)$$

is surjective.

Proof. For every representation X there is a natural isomorphism

$$\text{Hom}(P(i), X) \cong X_i, \quad f \mapsto f_i(e_i),$$

where e_i is the trivial path at i . Under this identification, the map g_* is exactly the component map $g_i : M_i \rightarrow N_i$. Since g is surjective, each component g_i is surjective. Hence g_* is surjective. $\square$

4. (20 points) Show that for any Q the representations $I(i)$, $i \in Q_0$ are indecomposable.

Proof. In the finite acyclic setting, the dual Yoneda isomorphism gives

$$\mathrm{Hom}(X, I(i)) \cong D(X_i)$$

naturally in X . Taking $X = I(i)$ yields

$$\mathrm{End}(I(i)) \cong D(I(i)_i).$$

Since the only path from i to i is the trivial path, $I(i)_i \cong k$, and therefore $\mathrm{End}(I(i)) \cong k$. If $I(i) = A \oplus B$ with both A and B nonzero, projection onto one summand would give a nonzero idempotent different from 1 in $\mathrm{End}(I(i))$, impossible in the field k . Thus $I(i)$ is indecomposable. $\square$

5. (20 points) Show that the smallest wild tree with valence at most 3 at each vertex has 7 vertices.

Proof. For acyclic quivers, representation type depends only on the underlying graph: Dynkin trees are finite type, Euclidean trees are tame, and all other connected trees are wild.

No tree with at most five vertices is wild: a path is of type A_n , and under the maximum-valence-3 condition the only branched possibilities are D_4 and D_5 . With six vertices and maximum valence at most 3, the possible trees are

$$A_6, \quad D_6, \quad E_6, \quad \tilde{D}_5,$$

which are Dynkin or Euclidean. Hence no such tree with at most six vertices is wild.

Now consider the seven-vertex tree with edges

$$ua, \quad ub, \quad uv, \quad vc, \quad vw, \quad wd.$$

It has maximum valence 3: the vertices u and v have valence 3, the vertex w has valence 2, and a, b, c, d are leaves. It is not Dynkin of type A, D, E , and it is not Euclidean: it is not a cycle, not one of the exceptional $\tilde{E}$ diagrams, and it is not a $\tilde{D}_n$ diagram because the two trivalent vertices are adjacent while one outer arm has length 2. Therefore it is wild. Thus the smallest possible number of vertices is 7. $\square$